

Course Code: BCE-C524
Course Name: CLOUD COMPUTING

MM: 100 Time: 3 Hr. L T P 3 1 0	Sessional:30 ESE:70 Credit :4
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Prerequisites:	Discrete Mathematics, Computer Networks
Objectives:	- To understand the concepts of Cloud Computing. - To learn Taxonomy of Virtualization Techniques. - To learn Cloud Computing Architecture. - To acquire knowledge on Aneka Cloud Application Platform. - To learn Industry Cloud Platforms.

NOTE:	The question paper shall consist of two sections A and B. Section A contains 10 short type questions of 6 marks each and student shall be required to attempt any five questions. Section B contains 8 long type questions of ten marks each and student shall be required to attempt any four questions. Questions shall be uniformly distributed from the entire syllabus.
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Module	Course Content	No. of Hours	POs mapped	PSOs mapped
Module-1	Overview of cloud computing: What is a cloud, Definition of cloud, Characteristics of cloud, why use clouds, How clouds are changing, Driving factors towards cloud, Comparing grid with cloud, Public clouds (commercial), Cloud Service Models (IaaS, PaaS, SaaS – Overview)	08	PO1	PSO1/ PSO2
Module-2	Cloud computing concepts: Concepts of cloud computing, Cloud computing leverages the Internet, Positioning cloud to a grid infrastructure, Elasticity and scalability, Virtualization, Characteristics of virtualization, Benefits of virtualization, Virtualization in cloud computing, Hypervisors, Multitenancy, Types of tenancy, Application programming interfaces (API), Billing and metering of services, Management, tooling, and automation in cloud computing.	08	PO1/ PO2	PSO1
Module-3	Cloud service delivery: Cloud service , Cloud service model architectures, Infrastructure as a service (IaaS) architecture , Infrastructure as a service (IaaS) details, Platform as a service (PaaS) architecture, Platform as a service (PaaS) details, Platform as a service (PaaS) , Examples of PaaS software, Software as a service (SaaS) architecture, Software as a service (SaaS) details, Function as a Service (FaaS) / Serverless Computing, Container as a Service (CaaS), AI as a Service (AIaaS) and ML as a Service (MLaaS), Backup as a Service (BaaS) , Examples of SaaS applications , Trade-off in cost to install versus , Common cloud management platform reference architecture: Architecture overview diagram, Common cloud management platform, Database as a Service - Monitoring as a Service – Communication as services.	08	PO2/ PO3	PSO1/ PSO2
Module-4	Cloud deployment scenarios: Cloud deployment models, Public clouds, Hybrid clouds, Community, Virtual private clouds, Vertical and special purpose, Migration paths for cloud, Selection criteria for cloud deployment, Case study example: AWS Platform.	08	PO3/ PO5/ PO12	PSO1/ PSO2

	Virtualization For Cloud Need for Virtualization – Pros and cons of Virtualization – Types of Virtualization – System Vm, Process VM, Virtual Machine monitor – Virtual machine properties - Interpretation and binary translation, HLL VM - Hypervisors – Xen, KVM , VMWare, Virtual Box, Hyper-V.			
Module-5	Security in cloud computing: Cloud security reference model, How security gets integrated, Cloud security, Understanding security risks, Principal security dangers to cloud computing, Data corruption or loss, User account and service hijacking, Steps to reduce cloud security breaches. Classification of Cloud Implementations: Amazon Web Services, The Elastic Compute Cloud (EC2). The Simple Storage Service (S3), AWS Lambda (Serverless), AWS RDS (Relational Database Service, The Simple Queuing Services (SQS), Google AppEngine - PaaS, Windows Azure; Aneka, Hadoop, IBM Cloud, Oracle Cloud, Alibaba Cloud, A Comparison of Cloud Computing Platforms.	08	PO1/ PO3	PSO1/ PSO2
Total No. of Hours		40		

Learning Outcomes:	- Understand the concept of virtualization and how this has enabled the development of Cloud Computing - Know the fundamentals of cloud, cloud Architectures and types of services in cloud - Understand scaling, cloud security and disaster management - Design different Applications in cloud - Ability to use AWS/IBM Cloud/Google cloud
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Suggested books:

S. No.	Name of Authors /Books /Publisher/Year
1.	Practices and Paradigms in Cloud Computing, RajKumarBuyya
2.	IBM , Handouts
3.	Michael Miller, Cloud Computing (1 ed.), Que Publishing, 2008. ISBN 978-0789738035.
4.	Cloud Computing, Publisher: Jones and Barret India, Author : Kris Jasm
5.	Anthony Velte, Toby Velte and Robert Elsenpeter, Cloud Computing: A practical Approach (1 ed.), Tata McGrawHill, 2009. ISBN 978-0070683518.

CO-PO/PSO MAPPING														
Course Outcomes (COs)	Program Outcomes (POs)												Program Specific Outcomes (PSOs)	
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	√	√											√	√
CO2	√	√	√										√	√
CO3	√	√	√										√	√
CO4	√	√										√	√	√
CO5	√	√	√		√							√	√	√